

Towards Safe Mid-Air Drone Interception: Strategies for Tracking & Capture

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Abstract—A unique approach for mid-air autonomous aerial interception of non-cooperating Uncrewed Aerial Vehicles by a flying robot equipped with a net is presented in this paper. A novel interception guidance method dubbed Fast Response Proportional Navigation (FRPN) is proposed, designed to catch agile maneuvering targets while relying on onboard state estimation and tracking. The proposed method is compared with state-of-the-art approaches in simulations using 100 different trajectories of the target with varying complexity comprising almost 14 h of flight data, and Fast Response Proportional Navigation (FRPN) demonstrates the shortest response time and the highest number of interceptions, which are key parameters of agile interception. To enable a robust transfer from theory and simulation to a real-world implementation, we aim to avoid overfitting to specific assumptions about the target, and to tackle interception of a target following an unknown general trajectory. Furthermore, we identify several often overlooked problems related to tracking and estimation of the target's state that can have a significant influence on the overall performance of the system. We propose the use of a novel state estimation filter based on the Interacting Multiple Model filter and a new measurement model. Simulated experiments show that the proposed solution provides significant improvements in estimation accuracy over the commonly employed Kalman Filter approaches when considering general trajectories. Based on these results, we employ the proposed filtering and guidance methods to implement a complete autonomous interception system, which is thoroughly evaluated in realistic simulations and tested in real-world experiments with a maneuvering target going far beyond the performance of any state-of-the-art solution.

Index Terms—Aerial systems; Perception and autonomy, field robots, reactive and sensor-based planning.

I. INTRODUCTION

INTERDICTION of Uncrewed Aerial Vehicles (UAVs) intruding in a restricted airspace is an important topic in aerial

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Video and data: <https://mrs.felk.cvut.cz/towards-interception>.

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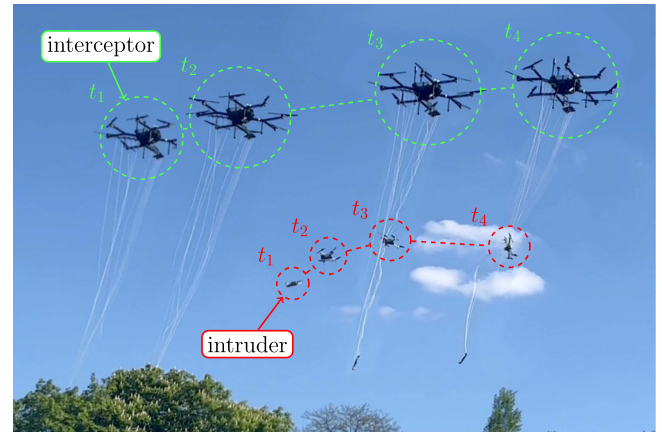


Fig. 1. Collage of a successful autonomous interception of a moving target using the proposed system. The maneuver took approx. 2 s from t_1 to t_4 .

safety, as evidenced by many recent UAV-related incidents and a growing research interest [1], [2], [3]. In this work, we aim to tackle autonomous aerial physical interception by a specialized robotic system with a catching net, as illustrated in Fig. 1. In comparison to other interdiction methods, which are mostly ground-based and/or manual, this approach offers several advantages, including the possibility of tracking and safely capturing fully autonomous and agile targets, the reduction of risk for collateral damage, and the lack of need for a skilled human operator in case of piloted systems. However, autonomous aerial interception also presents significant technical and research challenges. Some works have demonstrated an ability to capture a hovering target by a launched net [4], [5], but a fully working system that could reliably intercept fast maneuvering targets in real-world conditions has not yet been shown in the published literature. A net launcher only offers a limited amount of interception attempts and has also other limitations, such as the maximal flight velocity for a safe launch, which is why we choose to use a hanging net.

The complexity of the problem is illustrated by the MBZIRC 2020 robotic competition, where one of the challenges was inspired by the aerial interception problem¹. From 215 registered teams (22 expert international competitors selected for finals), only four teams were able to intercept the target, while relying on strong assumptions about the target's appearance and trajectory given in advance by the competition. Inspired by our own solution [6], which placed second in this challenge and first in the finals of the entire competition, we have developed a 3D Light

¹<https://mrs.felk.cvut.cz/mbzirc2020>

Detection and Ranging (LiDAR)-based detection system that removes some of these assumptions and provides accurate detection, localization, and multi-target tracking of flying objects in real-world scenarios, as described in our recent work [7]. In this paper, we build upon that onboard perception and propose a novel interception guidance technique. This, together with the proposed advanced state estimation and trajectory prediction of the target relying on minimal assumptions about the target's trajectory, enables us to reach a full system capable of real-world deployment.

A. Related Works

The two main research domains relevant to designing an Autonomous Aerial Interception System (AAIS) are detection and tracking of the target, and guidance of the interceptor towards the target. The detection and tracking of intruder UAVs typically include also estimation of the target's state that serves as input for the subsequent guidance algorithm, and thus forms the foundation of an effective interception. Assuming a well-estimated state of the target, the system's capability to intercept it is then determined by the navigation algorithm. An overview of related works in these two fields is provided in this section.

1) *Detection and Tracking*: Most of the research on onboard detection of non-cooperating UAVs focuses on visual methods based on deep learning [8], [9], [10]. However, detection and precise state estimation of small drones in cluttered environments proves to be a difficult problem, which several works attempt to address by utilizing depth images instead of (or in addition to) RGB [4], [11]. Flying objects are typically spatially distinctive, making them easily detectable in depth images, and the depth information provides a full 3D relative location of the target, which is otherwise problematic with monocular approaches. Our recent research [7] further builds upon this idea and takes advantage of the high accuracy of 3D LiDAR sensors to build a volumetric occupancy map of the environment and detect flying objects within this map, which provides more context and thus enables detection with a minimum of false positives even in complex environments.

Accurate state estimation and trajectory prediction of the target is crucial for agile interception. In [12], an approach for bearing-only position and velocity estimation of a visually-tracked target based on a modified Kalman Filter (KF) is proposed. If the full 3D position of the target is measured, a Constant Velocity (CV) or Constant Acceleration (CA) KF is typically employed, such as in [7], [8], [13]. However, we have observed that assuming a CV or even a CA dynamic model proves insufficient to predict trajectories of targets following non-trivial trajectories. In this work, we rely on the output of the detection and tracking system from [7], and go beyond that solution by integrating an Interacting Multiple Model (IMM) filter, which offers better accuracy in the considered cases, as confirmed by our experiments.

2) *Navigation for UAV Interception*: The simplest approach to interception guidance is Pure Pursuit (PP), which only considers the target's current relative position. This approach is used in [14], where a cooperative multi-UAV interception strategy is proposed. A more sophisticated method is Proportional Navigation (PN), which is a commonly used technique originally developed for missile guidance that generally provides more accurate interception than PP. In [15], PN-based guidance combined with a CA model of the target is designed and evaluated on three

trajectories in real-world experiments with a virtual UAV target. A major limitation of PN applied to UAVs, is identified in [16]. For the PN guidance law to converge towards interception, the relative velocity between the interceptor and the target must be negative. The authors propose a modification dubbed General Proportional Navigation (GPN) to address this limitation. The interception planning problem can also be formulated using Model Predictive Control (MPC) optimization as in [13], where the interceptor's desired acceleration and heading is selected by minimizing the Euclidean distance between the predicted trajectory of the target and the interceptor. However, the authors of [13] assume a car-like model of the interceptor UAV, which is overly restrictive and leads to a reduced performance. An often overlooked problem is a limited field of view of real sensors. This is tackled in [17], where a Reinforcement Learning (RL)-based planning strategy for tracking multiple targets using agents with limited field of view is presented. Although RL is a promising field in motion planning, it is yet to be applied to aerial interception.

Interception using a launched net is a similar task to the one in this paper, but it has different navigation requirements. This is tackled in [4] using PP, and in [5] using a custom guidance law that also takes into account velocity of the target. Both works demonstrate a real-world capture of a hovering target using these techniques while relying on feedback from onboard cameras, putting them the closest to a fully working AAIS. However, such navigation is not easily transferable to agile interception using a hanging net.

B. Contributions

In the context of the state of the art, we summarize our contributions as follows:

- 1) Two new methods for guiding an AAIS UAV to physically intercept another UAV are proposed: an MPC-based trajectory planner, and a PN-based Fast Response Proportional Navigation (FRPN) guidance law.
- 2) A detailed comparison of several approaches to state estimation of the target is provided, including a novel measurement uncertainty model.
- 3) The methods are integrated into a full AAIS solution and compared with state-of-the-art state estimation and guidance algorithms. The proposed methods offer a significant improvement in time to first contact with the target and in the number of successful interceptions.
- 4) Our dataset of testing trajectories is publicly available to enable reproducing our results and to enable comparison with future new interception navigation methods.
- 5) Finally, viability of the designed AAIS system is demonstrated in several real-world experiments, showing the capability to intercept a target flying 5 ms^{-1} .

II. STATE ESTIMATION AND PREDICTION

As mentioned in the previous sections, we rely on the LiDAR-based Volumetric Flying Object Detector (VoFOD) algorithm presented in [7] to detect the target UAV, and on the multi-target tracking algorithm from the same work. To estimate the state of the target and predict its motion, VoFOD relies on a standard KF and assumes a CA motion model. Although most typical UAV movement patterns consist of waypoint-following or similar trivial trajectories for which a well-tuned CV or CA model

is sufficient, in this work, we aim to design a solution that works more universally also for agile maneuvering targets. As the results in Section IV-A show, a suitably formulated and well-tuned IMM filter provides superior performance in this regard.

A. Interacting Multiple Models Filter

An IMM filter consists of m estimators running in parallel, each assuming a different dynamic model. Their estimates are combined based on the assumption that the transitions between the dynamic models follow a Markov chain model. This is done in two phases: a *filtering* phase, and a *mixing* phase.

The filter estimates the likelihood $\mu_{[k]}^j$ of each model. Here, $[k]$ denotes the time step and j the j -th model. In the *filtering* phase, the separate estimators are updated using their respective models and a measurement to obtain state estimates $\hat{\mathbf{x}}_{[k-]}^j$, and the corresponding covariances $\mathbf{P}_{[k-]}^j$. In the *mixing* phase, the models interact and are re-initialized and re-weighted. For the j -th filter, this is performed as

$$\hat{\mathbf{x}}_{[k]}^j = \sum_{i=1}^m \mu_{[k]}^{i|j} \hat{\mathbf{x}}_{[k-]}^i, \quad (1)$$

$$\mathbf{P}_{[k]}^j = \sum_{i=1}^m \mu_{[k]}^{i|j} \left(\mathbf{P}_{[k-]}^i + [\hat{\mathbf{x}}_{[k]}^i - \hat{\mathbf{x}}_{[k]}^j][\hat{\mathbf{x}}_{[k]}^i - \hat{\mathbf{x}}_{[k]}^j]^\top \right), \quad (2)$$

where $\hat{\mathbf{x}}_{[k]}^j$ is the state estimate after mixing, $\mathbf{P}_{[k]}^j$ is the corresponding covariance, and $\mu_{[k]}^{i|j}$ is the conditional probability of model i transitioning to model j . The transition probability $\mu_{[k]}^{i|j}$ is computed using the measurement innovations of the individual filters and parameters of the Markov chain model. The overall estimate $\hat{\mathbf{x}}_{[k]}$ and covariance $\mathbf{P}_{[k]}$ are then obtained using the same (1), (2) with $\mu_{[k]}^i$ substituted for $\mu_{[k]}^{i|j}$. For a more thorough description of IMM, we refer the reader to [18].

In this work, we propose using an IMM filter with two motion models, CV and CA, and a position measurement of the target, which is provided by VoFOD. The dynamic models and the measurement model are formulated as stochastic discrete state-space systems defined as

$$\mathbf{x}_{[k+1]} = \mathbf{A}_{[k]} \mathbf{x}_{[k]} + \boldsymbol{\xi}_{[k]}, \quad \boldsymbol{\xi}_{[k]} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}_{[k]}), \quad (3)$$

$$\mathbf{z}_{[k]} = \mathbf{H} \mathbf{x}_{[k]} + \boldsymbol{\zeta}_{[k]}, \quad \boldsymbol{\zeta}_{[k]} \sim \mathcal{N}(\mathbf{0}, \mathbf{Z}_{[k]}), \quad (4)$$

where $\mathbf{x}$ is a state vector, $\mathbf{A}$ is a state transition matrix, $\boldsymbol{\xi}$ is a Gaussian process noise with zero mean and a covariance matrix $\boldsymbol{\Xi}$, $\mathbf{z}$ is a measurement, $\mathbf{H}$ is a measurement matrix, and $\boldsymbol{\zeta}$ is a Gaussian measurement noise with covariance $\mathbf{Z}$. The lower index $[k]$ will further be omitted unless relevant for brevity. The specific CV and CA models are described below, and are updated in the *filtering* phase as standard KFs.

1) *Constant Velocity Dynamic Model*: For the CV motion model, the variables from (3) and (4) are defined as

$$\mathbf{x} = [\mathbf{p}^\top \quad \mathbf{v}^\top]^\top, \quad \mathbf{A} = \begin{bmatrix} \mathbf{I} & \Delta t \mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad (5)$$

$$\boldsymbol{\xi} = \mathbf{B} \mathbf{a}, \quad \mathbf{B} = \begin{bmatrix} \frac{1}{2} \Delta t^2 \mathbf{I} \\ \Delta t \mathbf{I} \end{bmatrix}, \quad (6)$$

$$\mathbf{a} \sim \mathcal{N}(\mathbf{0}, \sigma_a^2 \mathbf{I}), \quad \mathbf{H} = [\mathbf{I} \quad \mathbf{0}], \quad (7)$$

where $\mathbf{p} \in \mathbb{R}^3$ and $\mathbf{v} \in \mathbb{R}^3$ denote the target's position and velocity, respectively, $\mathbf{I} \in \mathbb{R}^{3 \times 3}$ is an identity matrix, $\mathbf{0} \in \mathbb{R}^{3 \times 3}$ is a zero matrix, $\Delta t \in \mathbb{R}$ is the time-step duration, $\mathbf{B} \in \mathbb{R}^{6 \times 3}$ is an input matrix, and $\mathbf{a} \in \mathbb{R}^3$ is an unknown acceleration of the target that is assumed to be a Gaussian random variable with zero mean and covariance $\sigma_a^2 \mathbf{I}$. The process noise covariance matrix is thus $\boldsymbol{\Xi} = \sigma_a^2 \mathbf{B} \mathbf{B}^\top$. The measurement covariance matrix $\mathbf{Z}$ is obtained using the method described in Section II-B.

2) *Constant Acceleration Dynamic Model*: The CA motion model is described similarly to the CV model as

$$\mathbf{x} = [\mathbf{p}^\top \quad \mathbf{v}^\top \quad \mathbf{a}^\top]^\top, \quad \mathbf{A} = \begin{bmatrix} \mathbf{I} & \Delta t \mathbf{I} & \frac{1}{2} \Delta t^2 \mathbf{I} \\ \mathbf{0} & \mathbf{I} & \Delta t \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad (8)$$

$$\boldsymbol{\xi} = \mathbf{B} \mathbf{j}, \quad \mathbf{B} = \begin{bmatrix} \frac{1}{6} \Delta t^3 \mathbf{I} \\ \frac{1}{2} \Delta t^2 \mathbf{I} \\ \Delta t \mathbf{I} \end{bmatrix}, \quad (9)$$

$$\mathbf{j} \sim \mathcal{N}(\mathbf{0}, \sigma_j^2 \mathbf{I}), \quad \mathbf{H} = [\mathbf{I} \quad \mathbf{0} \quad \mathbf{0}], \quad (10)$$

where $\mathbf{j} \in \mathbb{R}^3$ is an unknown jerk of the target that is assumed to be a Gaussian random variable with zero mean and covariance $\sigma_j^2 \mathbf{I}$. The process covariance is thus $\boldsymbol{\Xi} = \sigma_j^2 \mathbf{B} \mathbf{B}^\top$.

B. LiDAR Measurement Uncertainty

For an accurate estimation of the target's state, an appropriate model of the measurement uncertainty is crucial. In the used motion models, the measurement $\mathbf{z} \in \mathbb{R}^3$ is defined as

$$\mathbf{z} = \mathbf{p} + \boldsymbol{\zeta}, \quad (11)$$

where $\mathbf{p}$ is the target's actual position in a static world frame $\mathcal{W}$, and $\boldsymbol{\zeta}$ is noise. Following the results from [7], we assume that there are three major sources of uncertainty in $\mathbf{z}$:

- 1) the range of the target l_m measured by the LiDAR,
- 2) the observer's own pose used to transform the LiDAR points from the observer's frame $\mathcal{O}$ to $\mathcal{W}$, which we define as a transformation $\mathbf{T}_{\mathcal{O}}^{\mathcal{W}}$,
- 3) biased sampling of the measured points of the target, which is observed from one direction.

For the first two uncertainties, we adopt the model from [7] and approximate the result as an additive Gaussian noise $\boldsymbol{\zeta}_{12} \in \mathbb{R}^3$, defined as

$$\boldsymbol{\zeta}_{12} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\zeta}_{12}}), \quad (12)$$

where the covariance $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}_{12}} \in \mathbb{R}^{3 \times 3}$ is a function of the pose $\mathbf{T}_{\mathcal{O}}^{\mathcal{W}}$, the target's measured range l_m , and their uncertainties expressed as a covariance $\boldsymbol{\Sigma}_{\mathbf{T}}$ and standard deviation σ_l . Based on the results from [7], we propose to model the third uncertainty as an additive Gaussian noise $\boldsymbol{\zeta}_3 \in \mathbb{R}^3$, defined as

$$\boldsymbol{\zeta}_3 \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\zeta}_3}), \quad (13)$$

where $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}_3} \in \mathbb{R}^{3 \times 3}$ is the covariance matrix. The probability distribution of the total measurement noise $\boldsymbol{\zeta} = \boldsymbol{\zeta}_{12} + \boldsymbol{\zeta}_3$ of the target's position $\mathbf{z}$ is then

$$\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\zeta}_{12}} + \boldsymbol{\Sigma}_{\boldsymbol{\zeta}_3}). \quad (14)$$

In practice, the uncertainty σ_l of the range measurement l_m is provided by the sensor manufacturer, and the error caused by the sampling bias is typically chosen as $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}_3} = \sigma_{\boldsymbol{\zeta}_3}^2 \mathbf{I}$ with $\sigma_{\boldsymbol{\zeta}_3}$ as an empirically determined parameter. To select the observer's pose uncertainty covariance $\boldsymbol{\Sigma}_{\mathbf{T}}$, we assume that the translation and rotation errors are independent, and that $\boldsymbol{\Sigma}_{\mathbf{T}}$ can thus be

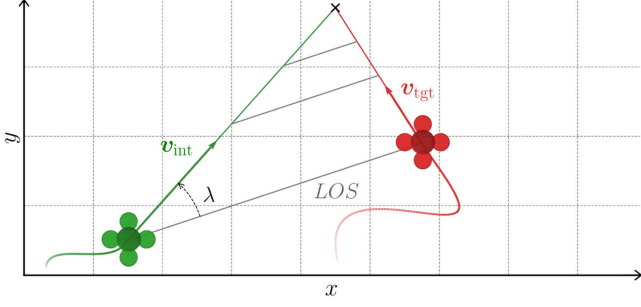


Fig. 2. Illustration of the fundamental concepts of Proportional Navigation (PN), highlighting the Line Of Sight (LOS). v_{int} denotes the velocity of the interceptor, and v_{tgt} denotes the velocity of the target.

expressed as

$$\Sigma_{\mathbf{T}} = \begin{bmatrix} \Sigma_t & \mathbf{0} \\ \mathbf{0} & \Sigma_\alpha \end{bmatrix}, \quad (15)$$

where $\Sigma_t \in \mathbb{R}^{3 \times 3}$ is the covariance of the translation error $\mathbf{t}_e \in \mathbb{R}^3$, and $\Sigma_\alpha \in \mathbb{R}^{3 \times 3}$ is the covariance of the rotation error $\alpha_e \in \mathbb{R}^3$, with $\alpha_e = [\alpha_e, \beta_e, \gamma_e]^\top$ representing rotations around the X , Y , and Z axes, respectively. It is important to note that the rotation uncertainty is expressed in the body frame of the observer, so the angles are assumed to be close to zero. The distribution is therefore linearized around $\alpha_e = \mathbf{0}$, so there is no singularity of the representation. We choose this model because it is intuitive to interpret and corresponds well to the observed behavior of the system.

The covariance Σ_t is reported by most self-localization systems, including the Real-time Kinematic (RTK) Global Positioning System (GPS) module used during our experiments. However, obtaining Σ_α is problematic, yet it can have a significant influence on the resulting total uncertainty. We observe, that for the widely adopted PixHawk 4 flight controller used in our system for attitude control and orientation estimation, Σ_α can be estimated as a function of angular velocities $\omega \in \mathbb{R}^3$ around the corresponding axes as

$$\Sigma_\alpha = c_\alpha^2 \text{diag}(\omega^2), \quad (16)$$

where c_α is an empirically determined parameter and ω^2 denotes an element-wise exponentiation of ω .

III. INTERCEPTION NAVIGATION AND PLANNING

In this section, we present the methods used for navigation of the interceptor towards interception of the target. Firstly, we discuss variants of Proportional Navigation (PN) that motivated the design of the proposed FRPN approach introduced in the second part. The last part presents a novel interception trajectory planner formulated using Model Predictive Control (MPC) reflecting the requirements of agile interception.

A. Proportional Navigation

Proportional Navigation (PN) is a guidance technique widely used in homing missile technology [19]. The central principle of PN involves the Line Of Sight (LOS), defined as a line connecting the interceptor and the target (see Fig. 2). If the interceptor and the target have a constant velocity and are on a collision course, the Line Of Sight (LOS) and the lines connecting the interceptor and the target with the collision point

form a so-called collision triangle. In such a case, it may be observed that the angle between the LOS and the interceptor's velocity is constant. This angle is commonly called the LOS angle and denoted λ . PN aims to reach this state by controlling the interceptor's acceleration to maintain a constant LOS angle.

The canonical PN control rule can be formulated as

$$\mathbf{a}_{\text{cmd}} = G \cdot v_c \cdot \dot{\lambda} \cdot \mathbf{a}_{\text{dir}}, \quad (17)$$

where $\mathbf{a}_{\text{cmd}} \in \mathbb{R}^3$ is the commanded acceleration, $G \in \mathbb{R}$ is control gain, $v_c \in \mathbb{R}$ is the closing velocity, $\dot{\lambda} \in \mathbb{R}$ is the rate of change of the LOS angle, and $\mathbf{a}_{\text{dir}} \in \mathbb{R}^3$ is a unit vector specifying the acceleration direction. The concrete definition of v_c and $\mathbf{a}_{\text{dir}}$ depends on the specific variation of PN used.

However as acknowledged in [16], this formulation has a major drawback in the tackled application. The LOS angle rate $\dot{\lambda}$ is zero not only when the target and the interceptor are flying along the sides of the collision triangle towards the collision point, but also when flying away from it along the same lines. Therefore, if the interceptor misses the target and continues flying, $\dot{\lambda}$ is close to zero, resulting in a very small $\mathbf{a}_{\text{cmd}}$, and thus a slow response of the interceptor. This issue is mostly irrelevant for missiles, and is therefore scarcely addressed in the literature, but it cannot be ignored for UAV interception, where the ability to reattempt the interception is crucial.

The problem can be circumvented by adopting an alternative PN formulation. Assuming a constant velocity of the target, unlimited action inputs of the interceptor, and several other conditions, PN can be derived using optimal control theory regarding the miss distance as the objective, yielding what is termed the Cartesian form or the Linearized Proportional Navigation (LPN) (for more details, we refer the reader to [20]). The LPN control rule can be formulated as

$$\mathbf{a}_{\text{cmd}} = G \frac{\Delta \mathbf{p} + \Delta \mathbf{v} \cdot t_{\text{go}}}{t_{\text{go}}^2}, \quad (18)$$

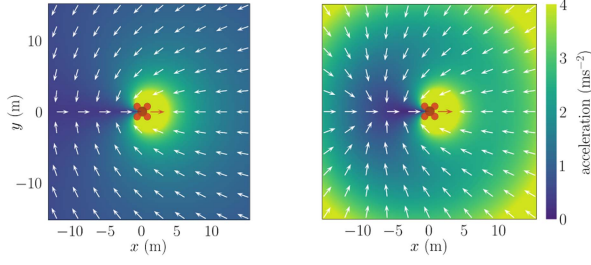
where $\Delta \mathbf{p} \in \mathbb{R}^3$ is the relative position of the target, $\Delta \mathbf{v} \in \mathbb{R}^3$ is the relative velocity, and $t_{\text{go}} \in \mathbb{R}$ is the time to collision estimated as $t_{\text{go}} = \frac{\|\Delta \mathbf{p}\|}{\|\Delta \mathbf{v}\|}$.

B. Fast Response Proportional Navigation Formulation

The above mentioned solution avoids the divergence problem, but fails to initiate engagement when $\Delta \mathbf{v}$ is close to zero. Moreover, as shown in Section IV-B, the engagement is slow and results in a long time to first contact. On the other hand, the simple PP control law produces an output proportional to the distance to the target resulting in a fast approach, but is prone to overshooting and large miss distances, which is why it is typically not employed. We propose a new control law, termed Fast Response Proportional Navigation (FRPN), to provide a solution suited for the agile interception of highly maneuvering targets. The proposed FRPN control rule is formulated as

$$\mathbf{a}_{\text{cmd}} = G \left((1 - W) \frac{\Delta \mathbf{p} + \Delta \mathbf{v} \cdot t_{\text{go}}}{t_{\text{go}}^2} + W \cdot \Delta \mathbf{p} \right), \quad (19)$$

where $W \in \mathbb{R}$ is a weight of the newly introduced proportional approach term $\Delta \mathbf{p}$. Fig. 3 compares the commanded accelerations of LPN and FRPN to illustrate the improved initial engagement of FRPN. As we demonstrate in Section IV-B, this modification significantly improves the speed of the initial and repeated engagements, which enables more attempts of the interceptor to catch the target, at a negligible cost in accuracy.



(a) Linearized Proportional Navigation. (b) Fast Response Proportional Navigation.

Fig. 3. Comparison between the commanded acceleration of LPN and FRPN. The white arrows indicate the acceleration and the color its magnitude for different positions of the interceptor. The target is positioned in the center and is flying at 1 ms^{-1} along the x -axis (red arrow). The interceptor is flying at 2 ms^{-1} in the same direction.

C. Model Predictive Control

Although Proportional Navigation can achieve optimality under specific conditions with respect to the miss distance, it cannot incorporate dynamic constraints of the UAVs. To overcome this limitation, we adopt the MPC framework and design a trajectory planning algorithm which implements a motion model of the target and the interceptor. We model both UAVs as point masses up to the second derivation similarly as the CA motion model presented in Section II-A2. Specifically, we denote the states of the interceptor and the target as

$$\mathbf{x}_{\text{int}} = [\mathbf{p}_{\text{int}}^T \quad \mathbf{v}_{\text{int}}^T \quad \mathbf{a}_{\text{int}}^T]^T, \quad (20)$$

$$\mathbf{x}_{\text{tgt}} = [\mathbf{p}_{\text{tgt}}^T \quad \mathbf{v}_{\text{tgt}}^T \quad \mathbf{a}_{\text{tgt}}^T]^T. \quad (21)$$

The interceptor's state $\mathbf{x}_{\text{int}} \in \mathbb{R}^9$ is produced by its self-localization and estimation pipeline using its onboard sensors, and the target's state $\mathbf{x}_{\text{tgt}} \in \mathbb{R}^9$ is estimated by the filtering algorithm, as discussed in Section II-A. This formulation allows for prediction of the interceptor's future state $\hat{\mathbf{x}}_{\text{int}}$ and the target's future state $\hat{\mathbf{x}}_{\text{tgt}}$ using the motion model defined in the (3), (8), and (9), by setting $j = 0$.

The objective of the MPC is to minimize the predicted position error $\hat{\mathbf{e}} = \hat{\mathbf{p}}_{\text{tgt}} - \hat{\mathbf{p}}_{\text{int}}$ between the interceptor and the target over a control horizon by controlling the input $\mathbf{u}$, which we define as $\mathbf{u} = \mathbf{a}_{\text{int}}$. The control horizon is discretized to N steps each with duration Δt , which we denote as $k \in \{0, 1, \dots, N-1\}$, with $k = 0$ corresponding to the current time t_0 , $k = 1$ to $t_0 + \Delta t$, etc. The optimal control problem is then formulated as

$$\mathcal{U}^* = \arg\min_{\mathcal{U}} J(\mathcal{U}), \quad (22)$$

$$J(\mathcal{U}) = \sum_{k=0}^{N-1} \hat{\mathbf{e}}_{[k]}^T \mathbf{W}_e \hat{\mathbf{e}}_{[k]} + \sum_{k=0}^{N-1} \mathbf{u}_{[k]}^T \mathbf{W}_u \mathbf{u}_{[k]}, \quad (23)$$

subject to

$$\hat{\mathbf{x}}_{\text{int}[0]} = \mathbf{x}_{\text{int}[0]}, \quad \hat{\mathbf{x}}_{\text{tgt}[0]} = \mathbf{x}_{\text{tgt}[0]}, \quad (24)$$

$$\hat{\mathbf{x}}_{\text{int}[k+1]} = \mathbf{A} \hat{\mathbf{x}}_{\text{int}[k]}, \quad \hat{\mathbf{x}}_{\text{tgt}[k+1]} = \mathbf{A} \hat{\mathbf{x}}_{\text{tgt}[k]}, \quad (25)$$

$$-\mathbf{v}_{\text{max}} \leq \mathbf{v}_{\text{int}[k]} \leq \mathbf{v}_{\text{max}}, \quad -\mathbf{a}_{\text{max}} \leq \mathbf{a}_{\text{int}[k]} \leq \mathbf{a}_{\text{max}}, \quad (26)$$

where $\mathcal{U} = \{\mathbf{u}_{[k]}\}$ is an input sequence of N control inputs, $\mathcal{U}^*$ is the optimal input sequence, $J(\mathcal{U})$ is the cost function, $\mathbf{W}_e \in \mathbb{R}^{3 \times 3}$ is an error weighting matrix, $\mathbf{W}_u \in \mathbb{R}^{3 \times 3}$ is an input weighting matrix, and $\mathbf{v}_{\text{max}}$ and $\mathbf{a}_{\text{max}}$ are the interceptor's

dynamic constraints. For the experimental evaluation, the optimization problem is implemented and solved using the ACADOS MPC framework [21].

IV. EXPERIMENTS

In this section, we validate our system in simulations by addressing the following questions:

- **Tracking a maneuvering target:** Is the IMM filter more effective than traditional approaches? What is the impact of agile maneuvering on the quality of estimation when utilizing the proposed measurement model?
- **Fast Response Proportional Navigation:** How does the proposed FRPN compare with established variants of PN in terms of effectiveness? How does our adjustment affect the time to first contact and interception accuracy?
- **Evaluation in realistic simulations:** How do the considered methods perform in realistic simulations that include a fully simulated Software In The Loop (SITL) UAV control pipeline, LiDAR-based detection, tracking, and target's state estimation?

Finally, we demonstrate our system's practicality in real-world experiments including an autonomous capture of a maneuvering target. The proposed methods are implemented using the MRS UAV system [22], which provides self-localization and state estimation, trajectory tracking, and low-level control for UAVs (among other sub-systems).

Unless specified otherwise, the experiments were simulated in the Gazebo simulator with the MRS X500 UAV (see [23]) for the interceptor and the target, which is a quadcopter based on the Holybro X500 frame with an approx. 0.5 m diameter, 3.7 thrust-to-weight ratio, and weights 3 kg (incl. the battery). The real-world experiments were realized using the custom EagleOne² interceptor, which is approx. 1.6 m in diameter, has a 3.1 thrust-to-weight ratio, weights 16 kg, and is equipped with the Ouster OS1-128 LiDAR, and a 2 m × 3 m deployable net (depicted in Fig. 1). Both UAV platforms carry an RTK-GPS for self-localization, an Intel NUC computer running the state estimation, control, detection, and navigation, and a PixHawk 4 flight controller for low-level stabilization and control. Videos from the experiments and the testing trajectories are available online³.

A. Tracking a Maneuvering Target

We compare the state estimation algorithms and evaluate the proposed LiDAR measurement uncertainty model on detections of the target using the VoFOD algorithm in a high-fidelity SITL Gazebo simulation environment that provides a realistic simulation of the LiDAR sensor. Six trajectories of various complexity are selected from the 100 testing trajectories designed for the quantitative evaluation of navigation methods (see the next section). Only a subset is selected for this evaluation because simulating the full set manually is infeasible (Gazebo does not provide an easy way to automate such task). For each trajectory tracked by the target, three sets of data are captured with the interceptor first *hovering*, then slowly *flying* (with body rotation rates $\leq 1 \text{ rads}^{-1}$), and then *maneuvering* (rotation rates up to 2.3 rads^{-1}). The target achieves velocities up to 8 ms^{-1}

²<https://eagle.one>

³<https://mrs.felk.cvut.cz/towards-interception>

TABLE I

COMPARISON OF THE STATE ESTIMATION METHODS WITH AND WITHOUT THE MEASUREMENT UNCERTAINTY MODEL FOR DIFFERENT MOVEMENT MODES OF THE INTERCEPTOR. FOR EACH METRIC AND MOVEMENT MODE, THE BEST RESULTS AMONG THE SIX CONSIDERED FILTER VARIANTS ARE HIGHLIGHTED. THE CORRESPONDING MAX. BODY ROTATION RATE (ω) AND ORIENTATION UNCERTAINTY (σ_α) FOR EACH MODE ARE LISTED IN THE BRACKETS. THE DISTANCE OF THE TARGET WAS BETWEEN 5 M AND 25 M

Interceptor mode	No measurement model			With measurement model		
	e_x (-)	e_p (m)	e_v (m/s)	e_x (-)	e_p (m)	e_v (m/s)
Hovering ($\omega \sim 0 \text{ rad s}^{-1}$, $\sigma_\alpha \sim 0 \text{ rad}$)						
CV	0.559	0.285	0.481	0.559	0.285	0.481
CA	0.540	0.289	0.457	0.539	0.288	0.455
IMM	0.515	0.285	0.429	0.517	0.285	0.431
Flying ($\omega \leq 1 \text{ rad s}^{-1}$, $\sigma_\alpha \leq 0.005 \text{ rad}$)						
CV	0.728	0.271	0.675	0.586	0.240	0.534
CA	0.693	0.275	0.636	0.573	0.250	0.516
IMM	0.651	0.270	0.592	0.531	0.238	0.475
Maneuvering ($\omega \leq 2.3 \text{ rad s}^{-1}$, $\sigma_\alpha \leq 0.011 \text{ rad}$)						
CV	0.919	0.326	0.859	0.670	0.260	0.617
CA	0.864	0.324	0.801	0.647	0.262	0.592
IMM	0.816	0.317	0.752	0.602	0.253	0.542

The bold value mark the best-performing filters in the given category.

and accelerations up to 4 ms^{-2} during these experiments. Each experiment lasts approx. 90 s, and the detections have a rate of 10 Hz, which results in almost 30 min of data and around 17000 detection samples in total.

The performance of the state estimators is significantly influenced by the selection of their parameters, and manual calibration of these methods is susceptible to bias and oversights. Although there exist methods for tuning KFs, systematic tuning of the IMM filter is not well-established. To provide a fair comparison of the different algorithms, we tune the filters through numerical optimization by minimizing the Root Mean Square Error (RMSE) over the measured data using the Broyden-Fletcher-Goldfarb-Shanno algorithm.

Specifically, the optimization metric is defined as

$$e_x = \sqrt{\frac{1}{n} \sum_{i=1}^n \left\| \mathbf{p}_i - \mathbf{p}_i^{\text{gt}} \right\|_2^2 + c_e^2 \left\| \mathbf{v}_i - \mathbf{v}_i^{\text{gt}} \right\|_2^2}, \quad (27)$$

where n is the total number of samples, $\mathbf{p}_i$, and $\mathbf{v}_i$ are the i -th estimated position and velocity of the target, $\mathbf{p}_i^{\text{gt}}$, and $\mathbf{v}_i^{\text{gt}}$ are the corresponding ground truth, and $c_e = 1 \text{ s}$ is a unit normalization constant. This metric is also used to compare the performance of the estimators. We also separately compare the position error e_p and the velocity error e_v (defined analogously to e_x). The results are presented in Table I.

It may be observed, that the IMM outperforms the simpler models in almost all cases, which may be expected as the trajectories generally do not have a constant acceleration or velocity. The results confirm a better generalization ability of the IMM, which is recognized for its efficacy in handling trajectories that undergo mode changes between CV and CA [18]. This is a behavior commonly observed in autonomous drones during scanning or observation tasks. For all these reasons, we conclude that IMM is highly suitable for the task of intercepting non-cooperative UAVs.

All of the methods exhibited decreased performance with more aggressive interceptor motion, which we attribute to worsened accuracy of the interceptor's self-localization and state

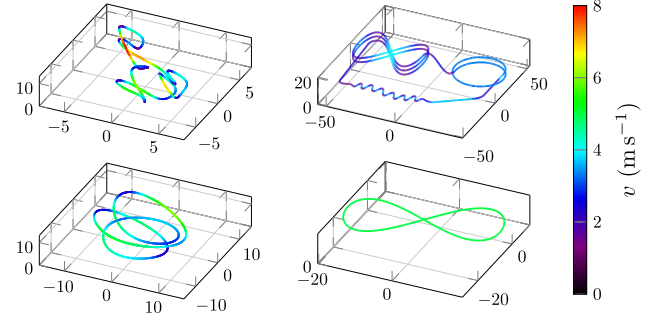


Fig. 4. Example trajectories of the target from the evaluation dataset (left column), the trajectory used in the full-system simulation (top right) and the fast trajectory used in the real-world experiments (bottom right). Color encodes velocity, dimensions are in meters.

estimation, as discussed in Section II-B. By incorporating the measurement uncertainty model described in that section, the increase in error is significantly mitigated. The only exception is for the *hovering* case where the uncertainty is negligible, so the results are similar with and without the model.

B. Quantitative Evaluation of the Navigation Methods

We compare the two proposed interception guidance methods with two state-of-the-art methods: the GPN control rule from [16], and the MPC formulation from [13], and two baseline methods: the PP and the LPN. To provide a robust comparison, the methods were evaluated using a set of 100 target's trajectories (see Fig. 4), with five random starting points of the interceptor per trajectory, and each evaluation lasting 100s, providing almost 14 h of total flight time. The trajectories differ in complexity with the fastest 5% having an average velocity of 5.9 ms^{-1} (with 8 ms^{-1} maximum) and acceleration of 3.2 ms^{-2} (11 ms^{-2} max.). The overall average velocity and acceleration are 4.1 ms^{-1} and 1.4 ms^{-2} . To enable reproducing the results, the dataset is available online⁴.

The tests were conducted in a custom simulator⁵ that supports automated repeated faster-than-real-time simulation with changing starting conditions, which was necessary to make the extensive quantitative evaluation technically feasible. The simulator is high-fidelity, incorporating aerodynamic drag and full UAV dynamics up to the dynamics of single motors, but does not provide simulation of sensors. The ground truth state of the target was used as the observation for the navigation algorithms, to allow an examination of each method's limits without favoring a specific detection or tracking approach.

During evaluation, we count the number of *successful interceptions*, which are defined as a passing of the target's center through a circular plane with a 2 m radius corresponding to the net suspended below the interceptor. Furthermore, we define the *interception accuracy* as the distance from the net's center during the passing. Parameter values for PP, LPN, GPN, and FRPN were selected using grid-search optimization by maximizing the number of trajectories with at least one successful interception as the primary metric, and the number of interceptions per trajectory as the secondary metric (when the primary metric reached the

⁴<https://mrs.felk.cvut.cz/towards-interception>

⁵https://github.com/ctu-mrs/mrs_multicopter_simulator

TABLE II

PARAMETER VALUES OF THE NAVIGATION ALGORITHMS IN THE EXPERIMENTS

Method	Parameters
PP	$G = 0.83$
LPN	$G = 19.7$
FRPN (ours)	$G = 19.7, W = 5.1 \cdot 10^{-2}$
GPN ₁ [16]	$k_1 = 40, k_2 = 1, v_r = -5 \text{ m s}^{-1}$
GPN ₂ [16]	$k_1 = 69.5, k_2 = 5.8, v_r = -6.6 \text{ m s}^{-1}$
MPC (ours)	$N = 40, \Delta t = 0.2 \text{ s}, \mathbf{W}_e = \mathbf{I}_{3 \times 3}, \mathbf{W}_u = 0.1 \cdot \mathbf{I}_{3 \times 3}$
MPC [13]	$N = 20, \Delta t = 0.1 \text{ s}, \mathbf{W}_e = \text{diag}(0.5, 0.5, 1, 1), \mathbf{W}_u = \text{diag}(0.8, 0.8, 0.8, 0.5)$
all	$\mathbf{v}_{\max} = [8 \text{ m s}^{-1} \ 8 \text{ m s}^{-1} \ 4 \text{ m s}^{-1}]^T, \mathbf{a}_{\max} = [4 \text{ m s}^{-2} \ 4 \text{ m s}^{-2} \ 2 \text{ m s}^{-2}]^T, \omega_{\text{heading}, \max} = 2 \text{ rad s}^{-1}$

TABLE III

AVERAGE RESULTS FROM 500 SIMULATED EXPERIMENTS ON 100 DIFFERENT TRAJECTORIES FOR EACH NAVIGATION METHOD: THE PERCENTAGE OF TRAJECTORIES WITH AT LEAST ONE INTERCEPTION ATTEMPT, THE AVERAGE NUMBER OF SUCCESSFUL INTERCEPTIONS PER TRAJECTORY, THE TIME TO THE FIRST ATTEMPT, THE INTERCEPTION ACCURACY PER ATTEMPT, AND THE INTERCEPTION ACCURACY OF THE FIRST ATTEMPT

Method	Trajs.	N. ints.	t to 1 st int.	Int. accur. all / 1 st
PP	72 %	4.40	32.22 s	1.19 m / 1.23 m
LPN	95 %	12.65	19.73 s	0.17 m / 0.03 m
MPC [13]	95 %	7.77	12.83 s	0.38 m / 0.39 m
GPN ₁ [16]	100 %	10.84	21.40 s	0.17 m / 0.25 m
GPN ₂ [16]	100 %	17.60	8.15 s	0.18 m / 0.34 m
MPC (ours)	98 %	16.55	11.98 s	0.38 m / 0.38 m
FRPN (ours)	100 %	24.38	5.93 s	0.16 m / 0.04 m

maximum). We test GPN both with the original values from [16] (denoted GPN₁) and with the tuned parameters (denoted GPN₂). Parameters of the MPC-based methods cannot be easily optimized in this manner. For the heading-vector MPC [13], the values from [13] were used except for the dynamic constraints, which were set to match the platform's actual constraints. For our MPC formulation, the parameters were empirically tuned. The parameter values are listed in Table II, and the results are summarized in Table III.

Based on these experiments, the FRPN and GPN₂ methods have a higher number of total attempts and a significantly lower time to the first attempt than the baseline solutions and the MPC-based methods. FRPN has a similar performance in interception accuracy to LPN, whereas GPN₂ performs slightly worse. The lower performance of the MPC methods indicates that distance to the target may not be the best optimization criterion as it leads to less aggressive behavior when the interceptor is lagging closely behind the target. The MPC planner from [13] is less aggressive and accurate due to the strong limitations enforced by the formulation of the control output. The main advantage of our approach compared to other methods is the notable improvement for both the reaction times and the number of repeated interceptions.

C. Validation With a Full System

To validate performance of the tuned methods when integrated into a full interception system including a real detector, the Gazebo simulator was utilized. The system consists of a simulated LiDAR, the detection and tracking of flying objects using VoFOD [7], the IMM filter with the LiDAR measurement uncertainty model, and an interception navigation algorithm. We consider the same navigation methods as in the previous section

TABLE IV

RESULTS FROM THE SITL-SIMULATED INTERCEPTION EXPERIMENTS USING THE WHOLE CONTROL, DETECTION, ESTIMATION, TRACKING, AND PLANNING PIPELINE: THE NUMBER OF SUCCESSFUL INTERCEPTIONS, THE AVERAGE INTERCEPTION ACCURACY, THE DETECTION RECALL, THE ESTIMATION ERRORS AND THE COMPUTATION DURATION

Method	Num. ints.	Int. accur.	Det. recall	e_x (-)	e_p (m)	e_v (m/s)	$\bar{t}_c$
MPC [13]	77	1.13 m	51 %	0.79	0.38	0.70	4.2 ms
GPN ₁ [16]	47	1.05 m	47 %	0.84	0.35	0.77	130 μ s
GPN ₂ [16]	189	0.79 m	48 %	0.71	0.32	0.64	130 μ s
MPC (ours)	188	0.84 m	41 %	0.85	0.40	0.75	5.1 ms
FRPN (ours)	230	0.73 m	47 %	0.77	0.35	0.69	130 μ s

(with the same parameter values) except the poorly-performing baseline methods PP and LPN. To address the limited field of view of the detector, when the tracking is lost, the planning is interrupted and the interceptor starts rotating in place to re-acquire the detection. The UAVs were controlled using the MRS UAV system [22], which allows for control based on desired acceleration or trajectory tracking.

A testing trajectory lasting 330 s was designed that incorporates various modes of target behavior to comprehensively assess capabilities of the system (see Fig. 4). Aggressiveness of the target is lower in comparison to the trajectories from previous section to reduce the number of detection losses (the average velocity and acceleration are 2.7 ms^{-2} and 0.7 ms^{-2} , the maxima are 4 ms^{-1} and 2.5 ms^{-2}). For each interception algorithm, the target executed the trajectory twice, resulting in approximately 11 min of data for each method. We use the evaluation metrics defined in the previous sections except for the time to the first attempt (due to the limited sample size). In addition, we evaluate the average computation time $\bar{t}_c$, and the detection recall as the percentage of LiDAR scans in which the target was detected. The results are summarized in Table IV.

The experiments show that our FRPN method outperforms the others in the considered metrics with GPN and the proposed MPC as the two next best methods, which is consistent with the results from Section IV-B. It is worth noting, that the PN-based methods have lower and deterministic computational demands, and depend on fewer parameters. Thus, they are suitable for real-time applications, and they can be easily tuned using optimization, as described in Section IV-B, which can lead to better results than manual tuning. On the other hand, the MPC model could be extended to consider obstacles in the environment and improve safety of the interception. However, optimization-based tuning and obstacle-aware formulation of the MPC are beyond the scope of this paper.

D. Real-World Testing

Based on the simulated results, we have selected the IMM filter with the measurement uncertainty model and the FRPN planner for the real-world evaluation. The target was periodically following an eight-shaped trajectory in a $16 \text{ m} \times 40 \text{ m}$ area with a constant velocity of 3 ms^{-1} , an average acceleration of 0.7 ms^{-2} , and a maximal acceleration of 1.9 ms^{-2} (see Fig. 4). The interceptor started in-air at a predefined position near the target's trajectory, and was fully autonomous, running the whole interception pipeline. To test the limits of the system, the experiment was repeated with an increased velocity of the target to 5 ms^{-1} (corresponding to 1.9 ms^{-2} average and 4.9 ms^{-2}

TABLE V
RESULTS FROM THE REAL-WORLD EXPERIMENTS USING FRPN WITH THE WHOLE PIPELINE COMPARED TO SIMULATED RESULTS OF THE OTHER METHODS WITH THE SAME STARTING CONDITIONS AND TARGET'S TRAJECTORY AS IN THE REAL-WORLD

Method	Target's speed: 3 m/s			Target's speed: 5 m/s		
	Num. ints.	Int. accur.	t to 1 st int.	Num. ints.	Int. accur.	t to 1 st int.
MPC (ours)*	7	0.82 m	9.82 s	2	0.78 m	11.32 s
GPN ₂ [16]*	11	0.68 m	7.82 s	3	1.39 m	6.34 s
FRPN (ours)*	13	0.74 m	5.34 s	4	0.99 m	4.52 s
FRPN (ours)	11	0.88 m	6.22 s	3	1.03 m	2.54 s

*simulated results

maximal acceleration). The real-world experiments were only conducted with FRPN due to technical limitations and safety, so the experiments were reproduced in simulations with the other methods to provide a fair comparison. The results are presented in Table V and confirm that our method is capable of intercepting a real-world target flying 5 ms⁻¹ and significantly outperforms the state of the art. We have also performed several experiments with a catching net deployed below the interceptor (see Fig. 1), demonstrating the ability of the whole AAIS platform to capture the target. All the experiments are illustrated in the attached video and online⁶.

V. CONCLUSION

Two new planning methods for the safe mid-air interception of intruding UAVs are proposed in this paper. The first, dubbed FRPN, is inspired by PN techniques used for missile guidance. The proposed FRPN aims to address the requirements of agile UAV-based interception. The second method is formulated as an MPC problem optimizing the distance to the target over a prediction horizon. The methods are compared with two baseline and two state-of-the-art planners, and the FRPN method shows notable improvements in time to first contact and interception accuracy. This is complemented with a thorough analysis of state estimation algorithms for target tracking, and a new model for measurement uncertainty that is suitable for sensors onboard a UAV, such as the LiDAR sensor considered in this work. As we show in the experiments, incorporating the measurement uncertainty model significantly improves estimation accuracy during agile maneuvers, which is crucial for effective aerial interception. Finally, the best-performing methods are combined with our previously published LiDAR-based onboard detector of flying objects into a first complete Autonomous Aerial Interception System that is capable of a real-world interception of agile maneuvering targets.

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